UNIVERSITY OF WESTMINSTER

INFORMATICS INSTITUTE OF TECHNOLOGY

TEA SENSE: A Spectroscopic System for Estimating Moisture and Quality Parameters in Black Tea

Final Year Project Proposal

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ABSTRACT

This project proposes the development and evaluation of a rapid, non-destructive method for estimating moisture content in tea using Near-Infrared (NIR) spectroscopy and Artificial Intelligence (AI) based modelling. Moisture content is a critical quality parameter in the tea industry, affecting product quality, shelf life, storage, processing efficiency, and market value. Conventional moisture testing methods can be time-consuming and require laboratory-based analysis.

The proposed approach will collect spectral data from tea samples using NIR spectroscopy and analyse the relationship between spectral signatures and moisture content. Reference moisture values will be obtained using the oven-drying method to constant mass.

The study will assess the feasibility of using NIR spectroscopy and AI-based modelling for rapid on-site moisture analysis in tea processing and quality control. The proposed approach may provide tea factories, estates, exporters, and other industry stakeholders with a faster and non-destructive alternative for preliminary moisture assessment, subject to the predictive performance achieved during the study.

Keywords: Near-Infrared Spectroscopy (NIR), Tea Moisture, Artificial Intelligence, Machine Learning, Spectral Preprocessing, Quality Control

DECLARATION AND GENERATIVE AI DISCLOSURE

Generative Artificial Intelligence (AI) tools were used during the preparation of this project proposal to support language refinement, content structuring, and the improvement of clarity and readability. The use of these tools was limited to assisting with the presentation and development of the written content.

All technical decisions, research objectives, methodologies, interpretations, and final content were reviewed, evaluated, and approved by the author. The author remains fully responsible for the accuracy, originality, integrity, and academic quality of the submitted work.

Any use of Generative AI has been undertaken in accordance with the relevant academic integrity guidelines and requirements of the institution.

Signature: Saad_____ Date: 29-08-2026

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TABLE OF CONTENTS

ABSTRACT 2
DECLARATION AND GENERATIVE AI DISCLOSURE 3
ACKNOWLEDGEMENTS 4
TABLE OF CONTENTS 5
LIST OF FIGURES 8
LIST OF TABLES 9
LIST OF ABBREVIATIONS 10
GLOSSARY 11
CHAPTER 1: INTRODUCTION 1
1.1 Chapter overview 1
1.2 Problem background 1
1.2.1 Research domain context 1
1.2.2 Specific technological context 2
1.2.3 Practical and real-world context 4
1.3 Problem definition 5
1.3.1 Problem statement 6
1.4 Research motivation 6
1.5 Research gap analysis 7
Table 1.1: Research Gap Summary 10
1.6 Contribution to the body of knowledge 12
1.6.1 Research-domain contribution 12
1.6.2 Problem-domain contribution 12
1.7 Research challenges 12
1.8 Research questions 13
RQ1 13
RQ2 13
RQ3 13
RQ4 13
1.9 Research aim 14
1.10 Research objectives 14
1.11 Scope, delimitations and assumptions 15
1.12 Expected outcomes and deliverables 16
1.13 Chapter summary 17
CHAPTER 2: LITERATURE REVIEW 18
2.1 Chapter overview 18
2.2 Review strategy and concept map 18
2.2.1 Literature search and selection strategy 18
2.2.2 Concept map 18
2.3 Problem-domain review 18
2.3.1 Domain task and decision context 18
2.3.2 Data and operational challenges 18
2.3.3 Standards, regulations and professional expectations 18

2.4 Technological review by implementation stage 19
2.4.1 [Pipeline stage 1 — e.g. data acquisition and curation] 19
2.4.2 [Pipeline stage 2 — e.g. pre-processing and quality control] 19
2.4.3 [Pipeline stage 3 — e.g. representation and feature extraction] 19
2.4.4 [Pipeline stage 4 — e.g. core modelling approach] 19
2.4.5 [Pipeline stage 5 — e.g. reliability, uncertainty or robustness] 19
2.4.6 [Pipeline stage 6 — e.g. interpretability or output presentation] 19
2.4.7 [Pipeline stage 7 — e.g. training strategy and hyperparameter optimisation] 19
2.5 Existing work: thematic analysis 19
2.6 Dataset review and comparison 20
2.7 Evaluation and benchmarking review 20
2.8 Synthesis and final research positioning 20
2.9 Proposed conceptual architecture 20
2.10 Chapter summary 20
CHAPTER 3: RESEARCH, TECHNICAL AND PROJECT METHODOLOGY 21
3.1 Chapter overview 21
3.2 Research methodology using Saunders' Research Onion 21
3.3 Research design and hypotheses 21
3.4 Proposed technical methodology and implementation pipeline 21
3.4.1 [Data ingestion and governance] 21
3.4.2 [Data quality assessment] 21
3.4.3 [Pre-processing] 21
3.4.4 [Baseline construction] 22
3.4.5 [Proposed model or system] 22
3.4.6 [Reliability / uncertainty / robustness component] 22
3.4.7 [Output, explanation or presentation component] 22
3.4.8 [Reproducibility and experiment tracking] 22
3.5 Requirements engineering 22
3.5.1 Stakeholder analysis 22
3.5.2 Requirement elicitation and evidence 22
3.5.3 Functional and non-functional requirements 22
3.6 System and software design 22
3.7 Data requirements and management plan 23
3.8 Development methodology 23
3.9 Evaluation methodology 23
3.9.1 Dataset splits and validation protocol 23
3.9.2 Baselines and ablation studies 23
3.9.3 Metrics and analysis 23
3.9.4 Research question to evaluation traceability 24
3.10 Project management methodology 24
3.10.1 Schedule and milestones 24
3.10.2 Deliverables and dates 24
3.11 Resources and new technical skills 24
3.12 Risk analysis and mitigation 24

3.13 Preliminary proof of concept and feasibility evidence 24
3.14 Chapter summary 25
CHAPTER 4: LEGAL, ETHICAL, SOCIAL AND PROFESSIONAL ISSUES 26
4.1 Chapter overview 26
4.2 Social issues 26
4.3 Legal issues 26
4.4 Ethical issues 26
4.5 Professional issues 26
4.6 Responsible use of Generative AI tools 26
4.6.1 Intended and permitted use 27
4.6.2 Prohibited or high-risk use 27
4.6.3 Verification and human accountability 27
4.6.4 Privacy, confidentiality and intellectual property 27
4.6.5 Transparency, disclosure and AI-use register 27
4.6.6 Generative AI declaration 27
4.6.7 Project-specific Generative AI risk assessment 27
4.7 Chapter summary 28
REFERENCES 29
APPENDICES 30
Appendix A: Detailed search strategy and screening log 30
Appendix B: Concept map 30
Appendix C: Literature survey matrix 30
Appendix D: Research gap analysis matrix 30
Appendix E: Dataset comparison matrix and data dictionary 30
Appendix F: Stakeholder onion, elicitation instruments and evidence 30
Appendix G: Full functional and non-functional requirements 30
Appendix H: Use cases, use-case descriptions and activity/sequence diagrams 30
Appendix I: Detailed architecture and data-flow diagrams 30
Appendix J: Detailed Gantt chart, backlog and milestone plan 31
Appendix K: Full risk register and issue log 31
Appendix L: Generative AI use declaration and AI-use register 31
Appendix M: Preliminary proof-of-concept screenshots, test logs and sample outputs 31
Appendix N: Ethics clearance material and data-management approvals 31

LIST OF FIGURES

[GUIDE NOTE — delete before submission] Insert a Table of Figures field in Word (References > Insert Table of Figures, caption label 'Figure').

LIST OF TABLES

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LIST OF ABBREVIATIONS

[GUIDE NOTE — delete before submission] Alphabetical. Include only abbreviations actually used, with one consistent expansion each.

GLOSSARY

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CHAPTER 1: INTRODUCTION

1.1 Chapter overview

This chapter introduces the research domain of rapid tea quality assessment using spectroscopy and data-driven analytical methods. It begins by discussing the importance of tea quality measurement and the limitations associated with conventional testing approaches. The chapter then introduces the technological context of Near-Infrared (NIR) spectroscopy, other spectroscopic techniques, chemometrics, and Artificial Intelligence (AI) for non-destructive analysis. Following this, the practical problem and research motivation are presented, together with the research gaps, expected contributions, research challenges, questions, aim, objectives, scope, and expected outcomes. The chapter establishes the context and direction of the proposed research and leads into Chapter 2, which will review the existing literature and identify the most appropriate technological and methodological approaches for the study.

1.2 Problem background

1.2.1 Research domain context

Tea is an important agricultural and commercial product whose quality is influenced by a range of physical, chemical, and safety-related parameters. Depending on the stage of production and the intended application, these parameters may include moisture content, polyphenolic compounds, caffeine, pigments, pesticide residues, and other chemical characteristics. Accurate measurement of such parameters is important for quality control, processing, storage, grading, safety, and commercial decision-making.

Sri Lanka has a significant presence in the global tea industry, with Ceylon Tea being exported to markets around the world. In 2025, Sri Lankan tea exports across bulk, packeted tea, tea bags, instant tea, and green tea categories totalled approximately 245,787 metric tonnes. The scale and commercial importance of the industry create a need for efficient and reliable methods of assessing tea quality throughout the value chain.

Many conventional analytical approaches used for tea quality measurement require laboratory facilities, trained personnel, sample preparation, chemical reagents, or extended processing

times. For example, moisture determination commonly relies on gravimetric procedures in which a sample is dried and its mass loss is measured. Such approaches can provide reference values but may not always be suitable when rapid or frequent measurements are required during industrial operations.

Similarly, the determination of chemical parameters such as total polyphenols may involve extraction procedures and laboratory-based spectrophotometric or chromatographic analysis. These methods are valuable for accurate reference measurements; however, their preparation requirements and analytical complexity can limit their suitability for rapid preliminary assessment in operational environments.

As the tea industry increasingly requires faster access to quality-related information, there is growing interest in analytical approaches that can reduce sample preparation, minimise destructive testing, and support more rapid measurement. Spectroscopic sensing technologies offer a potential alternative because they capture information about the interaction between electromagnetic radiation and the chemical composition of a sample. When combined with appropriate statistical or machine learning models, spectral information may be used to estimate selected physical or chemical properties.

The central research domain of this study is therefore the investigation of spectroscopy-based and AI-assisted approaches for rapid tea quality assessment. The initial focus of the research will be on moisture content, as it represents a measurable and practically important parameter for evaluating the feasibility of the proposed approach.

1.2.2 Specific technological context

Spectroscopy is an analytical technique that examines how electromagnetic radiation interacts with a material. Different materials and chemical compositions can produce different spectral responses, creating patterns that may contain information about the properties of the sample.

Near-Infrared (NIR) spectroscopy is particularly relevant to agricultural and food analysis because it can capture information associated with molecular bonds containing hydrogen, including O-H, C-H, and N-H bonds. These interactions make NIR spectroscopy potentially

useful for analysing organic materials and estimating characteristics associated with their chemical composition.

NIR spectroscopy has been investigated in tea research for applications including quality evaluation, classification, chemical composition analysis, and monitoring of processing conditions. Previous research has also explored the use of NIR combined with chemometric techniques to estimate components such as moisture and polyphenolic compounds. However, predictive performance can depend significantly on factors such as sample type, tea variety, processing stage, environmental conditions, instrument characteristics, and the quality of the reference measurements used to build the model.

Raw spectral data often contain noise, scattering effects, baseline variation, and redundant information. Therefore, spectral preprocessing and feature-selection methods are commonly applied before model development. Techniques such as Standard Normal Variate (SNV) can be used to reduce scattering-related variation, while Savitzky-Golay filtering or derivatives may assist in smoothing spectral signals and enhancing relevant spectral features.

Chemometric approaches such as Partial Least Squares Regression (PLSR) are commonly used to establish relationships between spectral data and laboratory-measured reference values. Feature-selection approaches, including Competitive Adaptive Reweighted Sampling (CARS), may also be investigated to identify wavelengths that contribute most strongly to a prediction model.

In addition to traditional chemometric methods, machine learning and Artificial Intelligence techniques may provide alternative approaches for analysing complex spectral datasets. Algorithms can potentially identify non-linear relationships between spectral characteristics and measured quality parameters. However, the suitability of any particular algorithm cannot be assumed before experimental evaluation.

The present research therefore does not commit at this stage to a single final sensing technology or predictive model. Instead, the study proposes an experimental evaluation of NIR spectroscopy as the primary technology, while allowing the literature review and preliminary experimentation to inform whether complementary spectroscopic techniques or additional sensing approaches should be investigated.

Ultraviolet (UV) spectroscopy and other analytical technologies may be considered during the broader investigation where justified by the specific parameter being measured. However, the primary research focus of the initial study will remain the feasibility of rapid moisture estimation using spectroscopy-based sensing.

1.2.3 Practical and real-world context

Tea production involves multiple stages at which quality-related information may be valuable. Tea estates, factories, processors, laboratories, exporters, brokers, and other stakeholders may require information about the characteristics of tea samples for quality control and decision-making.

In many conventional workflows, a sample may need to be collected and transferred to a laboratory before a measurement can be completed. Depending on the parameter and reference method, this may involve drying, extraction, chemical treatment, or specialised analytical equipment. Although laboratory-based methods remain essential for producing reliable reference measurements, they may not always provide immediate results for operational decisions.

For moisture measurement, conventional oven-drying procedures require a sample to be heated and repeatedly measured until a stable mass is obtained. This provides a reference value against which alternative sensing methods can be evaluated, but it can take substantially longer than a direct spectral measurement.

A rapid spectroscopy-based system would not necessarily replace laboratory testing immediately. Instead, it could potentially function as a preliminary or complementary measurement approach, subject to its demonstrated predictive performance. Such a system could support faster screening of samples and help identify cases where additional laboratory analysis is required.

However, deployment in real tea-industry environments introduces practical constraints. Samples may differ in particle size, moisture distribution, temperature, variety, origin, processing stage, and presentation. Field conditions may also introduce variations in lighting, environmental conditions, and instrument handling. These factors must be considered during

experimental design because a model that performs well under controlled laboratory conditions may not necessarily perform equally well in operational environments.

Therefore, this research proposes to evaluate the technology using real tea samples and practical testing conditions to determine whether the approach demonstrates sufficient feasibility for further development.

1.3 Problem definition

Tea quality assessment requires the measurement of multiple physical and chemical parameters, yet many established reference methods involve laboratory-based procedures that can be time-consuming, destructive, or dependent on specialised equipment and sample preparation.

Moisture content is one example of a parameter for which reliable reference methods exist but may require a significant amount of time before a result is obtained. This creates a potential gap between the need for rapid operational information and the time required for conventional laboratory analysis.

Spectroscopic technologies, particularly NIR spectroscopy, have demonstrated potential for rapid and non-destructive analysis of agricultural and food products, including tea. However, successful application depends on the development and validation of predictive models against reliable reference measurements. Spectral models can also be affected by sample variability, instrument characteristics, preprocessing choices, and environmental conditions.

Existing research demonstrates that spectroscopy and chemometrics can be applied to a range of tea quality parameters. However, there remains a practical need to evaluate how such approaches perform with real samples and under conditions relevant to the intended application environment.

The proposed research does not assume that a single technology will be capable of accurately measuring all tea quality and safety parameters. Instead, it seeks to investigate the feasibility of a spectroscopy-based framework, beginning with moisture detection as the primary use case.

The research will therefore focus on collecting spectral information from tea samples, obtaining corresponding reference moisture measurements, developing and evaluating predictive models, and comparing the predicted values against established reference values.

The study will initially be limited to moisture content. Other parameters, including polyphenolic compounds and selected safety-related measurements, may be identified as future extensions depending on the scientific feasibility established through the literature review and experimental findings.

1.3.1 Problem statement

There is a need to evaluate whether spectroscopy-based sensing combined with chemometric and AI-based modelling can provide a sufficiently accurate and rapid method for estimating moisture content in tea under practical testing conditions.

1.4 Research motivation

The motivation for this research is both practical and technological.

From a practical perspective, the tea industry requires efficient approaches to quality assessment. Faster access to information about important parameters could potentially support more responsive quality-control processes. While conventional laboratory methods remain important for accurate reference testing, there may be opportunities to complement these methods with faster, non-destructive sensing approaches.

From a technological perspective, advances in spectroscopy, portable sensors, computational processing, and machine learning have created opportunities to analyse complex physical and chemical characteristics without relying entirely on traditional laboratory workflows. NIR spectroscopy is particularly interesting because spectral measurements can be obtained rapidly and without necessarily destroying the sample.

The academic motivation of this research is to evaluate the relationship between tea spectral data and reference moisture measurements and to determine which preprocessing and modelling approaches provide the most useful predictive performance for the selected samples.

The study is also motivated by the possibility of establishing a broader research framework for tea quality analysis. If the initial investigation demonstrates promising results for moisture estimation, the methodology could later be investigated for additional parameters such as polyphenolic compounds or other quality-related characteristics. However, these potential extensions should be scientifically evaluated individually rather than assuming that one sensing technology can measure every parameter.

1.5 Research gap analysis

The literature indicates that spectroscopy, particularly Near-Infrared (NIR) spectroscopy, has considerable potential for rapid and non-destructive analysis of tea and other agricultural products. However, the existing research also demonstrates that the effectiveness of spectroscopy-based prediction depends on the selected spectral range, preprocessing methods, reference measurements, modelling techniques, sample diversity, and testing conditions. Therefore, the research gap addressed by this project is not the absence of spectroscopy in tea analysis, but the need to evaluate how effectively an appropriate spectroscopy and predictive-modelling workflow can estimate tea moisture under practical conditions.

Hazarika et al. (2017) identify NIR spectroscopy as a rapid, non-destructive and environmentally preferable alternative to conventional tea-quality assessment methods. Their review highlights the potential of NIR spectroscopy for quantitative and qualitative tea analysis and identifies chemometrics as an important component in interpreting spectral information. However, the review primarily establishes the potential of the technology rather than providing a validated implementation for rapid moisture assessment under a specific practical tea-processing environment.

Das et al. (2020) demonstrate that NIR reflectance spectroscopy combined with chemometric analysis can discriminate green tea samples based on gallic acid content. Their study used Principal Component Analysis (PCA) and Partial Least Squares Regression (PLSR), with preprocessing producing a substantial improvement in the Class Separability Index from 53.63 to 93.20. This demonstrates that spectral preprocessing can significantly influence the usefulness of NIR data. However, the study focused on gallic acid and green tea

discrimination rather than moisture estimation, indicating that the suitability of a spectral modelling approach must be evaluated separately for different tea parameters.

More recent research further supports the potential of NIR spectroscopy for tea analysis. *Revilla et al. (2024)* investigated the use of NIRS for estimating heavy-metal composition and identifying tea blends. The study analysed 322 samples and used techniques including Modified Partial Least Squares (MPLS), PCA, SIMCA and OPLS-DA. Although strong calibration and classification performance was achieved, validation performance was lower for some blend categories. The authors also noted that heavy metals do not have direct absorption peaks in the NIR region and that their prediction depends on relationships with other components within the tea matrix. This highlights an important limitation: high predictive performance does not necessarily mean that the target parameter has a direct spectral signature, and model performance must therefore be evaluated carefully against independent reference measurements.

Hazarika et al. (2024) provide further evidence that portable NIR spectroscopy can be applied to rapid tea-quality assessment. Their study investigated commercial Crush Tear Curl (CTC) black tea and used preprocessing methods including Standard Normal Variate (SNV), Multiplicative Scatter Correction (MSC), and first- and second-order derivatives. PLSR was combined with variable-selection methods including Competitive Adaptive Reweighted Sampling (CARS) and Random Frog (RF). An important finding was that models using a substantially reduced number of wavelengths could provide strong predictive performance. This suggests that variable selection may be relevant when developing a practical system intended for faster or more efficient spectral analysis.

The broader 2026 review by Qiu et al. identifies several unresolved challenges across spectroscopic agricultural-quality applications. These include poor model universality across varieties and geographical regions, environmental interference, high equipment costs, large volumes of spectral data, and the need for portable and integrated systems. The review also identifies preprocessing techniques such as SNV and Savitzky-Golay filtering, feature-selection approaches such as CARS, and predictive models including PLSR, Support Vector Machines, Artificial Neural Networks and deep-learning approaches as important components of modern spectroscopic analysis.

Based on these studies, four principal gaps are identified for the proposed research.

Gap 1: Limited practical validation of tea moisture estimation

Previous research demonstrates the potential of NIR spectroscopy for analysing several tea characteristics, including gallic acid, theaflavin, liquor quality, blends, and heavy metals. However, the proposed research focuses specifically on determining whether spectral measurements can be used to estimate **tea moisture content rapidly under practical testing conditions**.

Moisture is particularly relevant because conventional gravimetric or oven-drying methods require the sample to undergo a drying process before the moisture value can be determined. The proposed research will therefore investigate whether NIR measurements can provide a substantially faster estimation while maintaining an acceptable relationship with the reference measurements.

Gap 2: Effect of preprocessing and wavelength selection

The literature demonstrates that raw spectral data may not provide the best predictive performance without preprocessing. *Das et al. (2020)* demonstrated a significant improvement in spectral discrimination following preprocessing, while *Hazarika et al. (2024)* demonstrated the usefulness of preprocessing and wavelength-selection methods for tea-quality prediction.

However, the appropriate preprocessing and wavelength-selection strategy for moisture prediction cannot simply be transferred from another tea parameter.

The proposed research will therefore evaluate preprocessing approaches such as SNV and Savitzky-Golay filtering and investigate CARS or related feature-selection methods where appropriate.

Gap 3: Model performance and generalisation

The literature shows that spectroscopy-based models can achieve strong results during calibration while experiencing reduced performance during validation. *Revilla et al. (2024)*, for example, reported differences between calibration and validation performance, while *Qiu et al. (2026)* identifies model universality as a major challenge across spectroscopic agricultural applications.

This creates a need to evaluate the proposed moisture model using held-out samples rather than relying only on training or calibration performance.

The proposed research will therefore separate model development from final evaluation and use independent or held-out samples to assess predictive performance.

Gap 4: Practical feasibility of rapid measurement

Spectroscopic analysis has the potential to reduce measurement time and sample preparation; however, predictive accuracy alone does not establish practical feasibility. Factors such as sample variability, measurement conditions, spectral noise, instrument characteristics and environmental variation can influence performance.

The proposed research will therefore investigate not only predictive performance but also the feasibility of using the developed approach for rapid tea-moisture assessment under selected practical or field conditions.

Table 1.1: Research Gap Summary

Research Gap	Evidence from Literature	Planned Research Response	Linked RQ	Linked Objective
Limited practical validation for moisture	NIR has been applied to several tea parameters,	Evaluate NIR specifically for tea moisture estimation	RQ1	O1, O2

	but performance is parameter-specific			
Preprocessing affects spectral performance	Das et al. (2020) demonstrated significant improvement following preprocessing	Compare SNV and Savitzky-Golay-based preprocessing	RQ2	O3
Importance of wavelength selection	Hazarika et al. (2024) demonstrated benefits from CARS and reduced spectral variables	Investigate CARS or related wavelength-selection methods	RQ3	O4
Calibration does not guarantee generalisation	Revilla et al. (2024) and Qiu et al. (2026) highlight validation and universality challenges	Evaluate models using held-out samples	RQ3	O5
Practical deployment challenges	Qiu et al. (2026) identifies environmental interference and model universality as challenges	Conduct practical/field feasibility testing	RQ4	O6
Need for rapid assessment	Existing methods may require laboratory analysis and preparation	Compare proposed spectral workflow with reference measurement workflow	RQ4	O6

1.6 Contribution to the body of knowledge

1.6.1 Research-domain contribution

The primary research contribution will be an experimental evaluation of a spectroscopy-based workflow for estimating moisture content in tea. Rather than proposing NIR spectroscopy as an entirely new technology for tea analysis, the research will investigate how spectral preprocessing, wavelength selection, and predictive modelling influence moisture-estimation performance for the selected tea samples. The study will provide comparative evidence on the suitability of approaches such as SNV, Savitzky-Golay filtering, PLSR and CARS for the specific prediction task. Performance will be evaluated against laboratory-derived reference values using held-out samples.

1.6.2 Problem-domain contribution

The practical contribution of the research will be an assessment of whether spectroscopy can support faster preliminary moisture assessment in a tea-industry context. The research will investigate the relationship between measurement speed, predictive performance and practical testing conditions rather than assuming that spectroscopy can immediately replace established laboratory methods. The findings will identify both the potential benefits and limitations of the proposed approach and may provide a foundation for future investigation of additional tea-quality parameters.

1.7 Research challenges

The proposed research presents several challenges that are important to the validity of the investigation.

Sample variability is a major challenge because tea samples may differ in variety, processing method, origin, particle size, storage conditions and moisture distribution. These differences may affect the resulting spectral patterns and consequently the predictive model.

Reference measurement consistency is also critical. Since the proposed model will learn the relationship between spectral measurements and moisture values, inaccurate or inconsistent reference measurements could directly affect model performance. Reference moisture values

will therefore be obtained using an established oven-drying method to provide the target values for model development and evaluation.

Spectral variability and noise may arise from scattering effects, sample presentation, instrument characteristics and environmental conditions. Appropriate preprocessing will therefore be investigated rather than assuming that raw spectra are directly suitable for modelling.

1.8 Research questions

RQ1

To what extent can NIR spectral measurements predict the moisture content of selected tea samples when compared with oven-drying reference measurements?

RQ2

How do spectral preprocessing techniques, specifically Standard Normal Variate (SNV) and Savitzky-Golay filtering, affect the predictive performance of tea moisture models?

RQ3

How do selected predictive modelling and wavelength-selection approaches affect the performance of NIR-based tea moisture estimation?

RQ4

To what extent does a NIR-based moisture estimation approach demonstrate feasibility for rapid measurement under selected practical or field conditions?

1.9 Research aim

The aim of this research is **to design, develop and evaluate a NIR spectroscopy-based approach for the rapid estimation of moisture content in tea using spectral preprocessing, wavelength selection and predictive modelling.**

The study will establish reference moisture values using an oven-drying method and collect corresponding NIR spectral measurements from selected tea samples. Different preprocessing and modelling approaches will then be investigated to determine their effect on predictive performance. The resulting models will be evaluated using held-out samples and appropriate statistical metrics. Finally, the research will assess whether the proposed approach demonstrates sufficient feasibility for rapid measurement under practical or field conditions.

1.10 Research objectives

The research will achieve the following objectives:

1. To review and critically evaluate existing research on NIR spectroscopy, chemometrics and machine learning for tea-quality and moisture analysis.

2. To collect NIR spectral measurements and corresponding reference moisture values from a selected range of tea samples.

3. To investigate the effect of spectral preprocessing techniques, including Standard Normal Variate (SNV) and Savitzky-Golay filtering, on moisture-prediction performance.

4. To investigate wavelength-selection techniques, including Competitive Adaptive Reweighted Sampling (CARS), for identifying informative spectral variables for moisture estimation.

5. To develop and compare selected predictive models, including Partial Least Squares Regression (PLSR) and appropriate machine-learning approaches, for estimating tea moisture content.

6. To evaluate model performance using held-out samples and metrics including Root Mean Square Error of Prediction (RMSEP) and coefficient of determination (R^2).

7. To assess the practical feasibility of using the developed spectroscopy-based approach for rapid tea moisture estimation under selected field or operational conditions.

1.11 Scope, delimitations and assumptions

The primary scope of this research is the investigation of NIR spectroscopy for estimating moisture content in tea. The research will involve collecting spectral data from selected tea samples and establishing corresponding reference moisture values using an oven-drying method.

The study will investigate spectral preprocessing, wavelength selection and predictive modelling. SNV, Savitzky-Golay filtering, PLSR and CARS will form part of the initial methodological investigation because these techniques have demonstrated relevance in the existing literature. Additional machine-learning techniques may be evaluated if the dataset and research findings justify their inclusion.

The study will include practical or field testing to assess whether the proposed approach can operate outside a strictly controlled laboratory environment. However, field testing will be treated as a feasibility assessment and not as evidence that the system is immediately suitable for commercial deployment.

The primary parameter is moisture content. Other parameters, including polyphenols, glyphosate, heavy metals and other tea-quality or safety characteristics, are not guaranteed to be included in the experimental phase. They may be considered as potential future extensions depending on the findings, available reference methods, equipment and research resources.

The research does not aim to replace established laboratory reference methods. Instead, the reference method will provide the ground-truth measurements required to evaluate the spectroscopy-based predictions.

The study assumes that the selected reference method provides sufficiently reliable moisture measurements and that the collected samples contain enough variation in moisture content to support meaningful statistical modelling.

The generalisability of the findings will be limited by the number and diversity of samples, tea types, processing conditions, instrument characteristics and field conditions included in the study.

1.12 Expected outcomes and deliverables

The expected outcomes of the research are:

Deliverable	Expected Output	Linked Objective
Literature Review	Critical review of spectroscopy and AI methods for tea-quality analysis	O1
Tea Sample Dataset	Documented tea samples representing selected conditions	O2
Reference Moisture Dataset	Oven-drying reference values for corresponding samples	O2
NIR Spectral Dataset	Spectral measurements associated with reference values	O2
Preprocessing Evaluation	Comparison of SNV and Savitzky-Golay preprocessing approaches	O3
Wavelength Selection	Evaluation of CARS or another suitable selection method	O4
Predictive Models	PLSR and selected machine-learning models	O5
Model Evaluation	RMSEP, R^2 and other relevant evaluation results using held-out samples	O6

Field Feasibility Assessment	Evaluation of measurement practicality and observed limitations	O7
Research Findings	Evidence-based conclusion regarding the feasibility of rapid NIR-based moisture estimation	O6, O7

1.13 Chapter summary

This chapter established the research context for investigating rapid, non-destructive tea moisture estimation using NIR spectroscopy and predictive modelling. Existing research demonstrates that NIR spectroscopy has considerable potential for tea-quality analysis, while also showing that preprocessing, wavelength selection, model choice, sample variability and validation have substantial effects on performance. Based on the literature reviewed, the research gap is therefore defined around the practical evaluation of a spectroscopy-based moisture-estimation workflow rather than the introduction of NIR spectroscopy as a new technology for tea analysis.

The proposed research will use oven-drying measurements as reference values and investigate NIR spectral preprocessing, wavelength selection and predictive modelling using techniques including SNV, Savitzky-Golay filtering, PLSR and CARS. Predictive performance will be evaluated using held-out samples and statistical measures including RMSEP and R^2. The research will also investigate practical or field feasibility to determine the limitations and potential of rapid spectroscopy-based moisture assessment. Chapter 2 will provide a detailed review of the relevant literature and establish the theoretical and technical basis for the experimental methodology.

CHAPTER 2: LITERATURE REVIEW

[GUIDE NOTE — delete before submission] Recommended length 13-14 pages. Critical and thematic. Not a sequence of one-paper-per-paragraph summaries.

2.1 Chapter overview

This chapter critically reviews existing research relevant to the proposed investigation of spectroscopy-based tea quality assessment, with particular emphasis on Near-Infrared (NIR) spectroscopy and its application to moisture estimation. The review is organised thematically rather than as a sequence of individual paper summaries. It first examines conventional approaches to tea quality assessment and their practical limitations, followed by the principles and applications of spectroscopy in agricultural and tea analysis. The review then considers NIR spectroscopy specifically, including its applications to tea quality parameters, spectral preprocessing, chemometric techniques, wavelength selection, and machine-learning approaches. The chapter subsequently examines reference measurement methods, model validation and generalisation, and the practical challenges associated with transferring spectroscopy-based models from controlled laboratory environments to field or industrial conditions. Finally, the reviewed evidence is synthesised to establish the methodological direction of the proposed research and to justify the focus on rapid tea moisture estimation.

2.2 Review strategy and concept map

2.2.1 Literature search and selection strategy

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2.2.2 Concept map

[GUIDE NOTE — delete before submission] One original figure plus a concise interpretation. Larger version in Appendix B if needed.

[Write your own content here.]

2.3 Problem-domain review

2.3.1 Domain task and decision context

[GUIDE NOTE — delete before submission] About one page. The target task, available inputs, the meaning of errors, and the role of human review.

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2.3.2 Data and operational challenges

[GUIDE NOTE — delete before submission] About one page. Explain the methodological implications of each data problem, not just its existence.

[Write your own content here.]

2.3.3 Standards, regulations and professional expectations

[GUIDE NOTE — delete before submission] About half a page. Apply them to this project rather than quoting legislation.

[Write your own content here.]

2.4 Technological review by implementation stage

[GUIDE NOTE — delete before submission] Follow your own proposed pipeline. Rename, merge or add subsections below so they match your project. For each stage: compare realistic alternatives, state selection criteria, and explain the effect on later stages. Roughly half to three-quarters of a page each.

2.4.1 [Pipeline stage 1 — e.g. data acquisition and curation]

[Write your own content here.]

2.4.2 [Pipeline stage 2 — e.g. pre-processing and quality control]

[Write your own content here.]

2.4.3 [Pipeline stage 3 — e.g. representation and feature extraction]

[Write your own content here.]

2.4.4 [Pipeline stage 4 — e.g. core modelling approach]

[Write your own content here.]

2.4.5 [Pipeline stage 5 — e.g. reliability, uncertainty or robustness]

[Write your own content here.]

2.4.6 [Pipeline stage 6 — e.g. interpretability or output presentation]

[Write your own content here.]

2.4.7 [Pipeline stage 7 — e.g. training strategy and hyperparameter optimisation]

[Write your own content here.]

2.5 Existing work: thematic analysis

[GUIDE NOTE — delete before submission] About three to four pages. Define three to five themes. Within each, compare methods, datasets and results, identify contradictions, and close with the implication for your work. Full literature survey matrix in Appendix C.

[Write your own content here.]

2.6 Dataset review and comparison

[GUIDE NOTE — delete before submission] About one page plus a comparison table: source, access, subjects, scans or records, structure, labels, class distribution, licence, ethics, limitations, planned role.

[Write your own content here.]

2.7 Evaluation and benchmarking review

[GUIDE NOTE — delete before submission] About one and a half pages. Splitting protocols, baseline selection, performance metrics, reliability metrics, robustness, efficiency, statistical uncertainty.

[Write your own content here.]

2.8 Synthesis and final research positioning

[GUIDE NOTE — delete before submission] About one page. What is established, what is unresolved, why your methods are appropriate, and how your work differs from the closest studies.

[Write your own content here.]

2.9 Proposed conceptual architecture

[GUIDE NOTE — delete before submission] One original figure plus about half a page of interpretation. Detailed version in Appendix I.

[Write your own content here.]

2.10 Chapter summary

[GUIDE NOTE — delete before submission] One paragraph. No new papers or methods.

[Write your own content here.]

CHAPTER 3: RESEARCH, TECHNICAL AND PROJECT METHODOLOGY

[GUIDE NOTE — delete before submission] Recommended length 9-10 pages. How the questions will be answered, how the artefact will be built, how evidence will be generated, and how feasibility and risk will be controlled.

3.1 Chapter overview

[GUIDE NOTE — delete before submission] One paragraph outlining the sequence of the chapter.

[Write your own content here.]

3.2 Research methodology using Saunders' Research Onion

[GUIDE NOTE — delete before submission] About one page with a justification table: philosophy, approach, methodological choice, strategy, time horizon, techniques. Justify each layer against your actual research questions, not as textbook theory.

[Write your own content here.]

3.3 Research design and hypotheses

[GUIDE NOTE — delete before submission] Half to three-quarters of a page. Independent and dependent variables, baselines and ablations, hypotheses, control of confounding factors.

[Write your own content here.]

3.4 Proposed technical methodology and implementation pipeline

[GUIDE NOTE — delete before submission] For every stage state inputs, processing, alternatives considered, selected approach, outputs, how it will be tested, and the fallback if it fails. Rename these subsections to match your own pipeline.

3.4.1 [Data ingestion and governance]

[Write your own content here.]

3.4.2 [Data quality assessment]

[Write your own content here.]

3.4.3 [Pre-processing]

[Write your own content here.]

3.4.4 [Baseline construction]

[Write your own content here.]

3.4.5 [Proposed model or system]

[Write your own content here.]

3.4.6 [Reliability / uncertainty / robustness component]

[Write your own content here.]

3.4.7 [Output, explanation or presentation component]

[Write your own content here.]

3.4.8 [Reproducibility and experiment tracking]

[Write your own content here.]

3.5 Requirements engineering

3.5.1 Stakeholder analysis

[GUIDE NOTE — delete before submission] One diagram plus a short interpretation. Analyse interest, influence and risk, not just a list of names.

[Write your own content here.]

3.5.2 Requirement elicitation and evidence

[GUIDE NOTE — delete before submission] Short subsection. How requirements were derived and validated, with traceability to the problem and objectives. Any human participation requires prior ethics approval.

[Write your own content here.]

3.5.3 Functional and non-functional requirements

[GUIDE NOTE — delete before submission] One concise MoSCoW table here: ID | Priority | Requirement | Verification method | Traceability. Full specification in Appendix G.

[Write your own content here.]

3.6 System and software design

[GUIDE NOTE — delete before submission] About one page here, with detailed diagrams in Appendices H and I. Context diagram, use cases, component or deployment architecture, data flow or sequence diagrams, interface wireframes.

[Write your own content here.]

3.7 Data requirements and management plan

[GUIDE NOTE — delete before submission] About half a page plus a dataset table. Roles, schema, train-validation-test strategy, subject-level separation, storage, backup, versioning, retention, licensing and ethics.

[Write your own content here.]

3.8 Development methodology

[GUIDE NOTE — delete before submission] About half a page. Selected approach, comparison with alternatives, iteration structure, definition of done, testing and change control.

[Write your own content here.]

3.9 Evaluation methodology

[GUIDE NOTE — delete before submission] This is the section examiners look at hardest. It must show exactly how each research question will be tested.

3.9.1 Dataset splits and validation protocol

[GUIDE NOTE — delete before submission] Short but exact. Leakage controls, cross-validation or hold-out rationale, no tuning on test data.

[Write your own content here.]

3.9.2 Baselines and ablation studies

[GUIDE NOTE — delete before submission] Simple baseline, strong baseline, and an ablation for each proposed component under a fair training budget.

[Write your own content here.]

3.9.3 Metrics and analysis

[GUIDE NOTE — delete before submission] Half to three-quarters of a page. Define metrics before experimentation and state what each one tests.

[Write your own content here.]

3.9.4 Research question to evaluation traceability

[GUIDE NOTE — delete before submission] Table: RQ | Experiment | Evidence | Decision supported.

[Write your own content here.]

3.10 Project management methodology

[GUIDE NOTE — delete before submission] About half a page. Approach, work packages, milestones, backlog and priority rules, progress review, issue log, supervisor meeting records.

[Write your own content here.]

3.10.1 Schedule and milestones

[GUIDE NOTE — delete before submission] Summary Gantt chart with dependencies, milestones and contingency here; detailed task-level chart in Appendix J.

[Write your own content here.]

3.10.2 Deliverables and dates

[GUIDE NOTE — delete before submission] One compact table: deliverable | linked objective | due date | acceptance criterion | dependency.

[Write your own content here.]

3.11 Resources and new technical skills

[GUIDE NOTE — delete before submission] About half a page. Hardware and compute, software, storage, domain expertise, new skills and how they will be acquired, access constraints.

[Write your own content here.]

3.12 Risk analysis and mitigation

[GUIDE NOTE — delete before submission] One table of the highest-priority risks: description | likelihood | impact | trigger | prevention | contingency | owner. Avoid generic risks such as 'time management'.

[Write your own content here.]

3.13 Preliminary proof of concept and feasibility evidence

[GUIDE NOTE — delete before submission] About half a page, with screenshots and logs in Appendix M. Report accurately and label clearly as preliminary.

[Write your own content here.]

3.14 Chapter summary

[GUIDE NOTE — delete before submission] One paragraph. Introduce no new methodological decisions.

[Write your own content here.]

CHAPTER 4: LEGAL, ETHICAL, SOCIAL AND PROFESSIONAL ISSUES

[GUIDE NOTE — delete before submission] Recommended length 2-3 pages. Project-specific, risk-based and action-oriented. For each issue: the risk, affected stakeholder, likely consequence, preventive control, contingency, and evidence the control will be followed.

4.1 Chapter overview

[GUIDE NOTE — delete before submission] One short paragraph naming the major themes and stating that mitigation is embedded in the methodology.

[Write your own content here.]

4.2 Social issues

[GUIDE NOTE — delete before submission] About half a page. Representation and unequal performance, accessibility, over-reliance and automation bias, who may be excluded.

[Write your own content here.]

4.3 Legal issues

[GUIDE NOTE — delete before submission] About half a page. Data protection, consent and anonymisation, dataset and software licences, copyright, third-party tool terms, cross-border storage.

[Write your own content here.]

4.4 Ethical issues

[GUIDE NOTE — delete before submission] About half a page. Ethics approval, data minimisation, bias and fairness, transparency, human oversight, error communication, non-fabrication of results.

[Write your own content here.]

4.5 Professional issues

[GUIDE NOTE — delete before submission] About half a page. Competence and limitations, accurate and reproducible reporting, secure development, responsible disclosure, documentation, professional codes.

[Write your own content here.]

4.6 Responsible use of Generative AI tools

4.6.1 Intended and permitted use

[GUIDE NOTE — delete before submission] State only the limited, transparent purposes actually used on this project.

[Write your own content here.]

4.6.2 Prohibited or high-risk use

[GUIDE NOTE — delete before submission] Fabricated citations, generated results, confidential inputs, disguised plagiarism, unverified domain claims.

[Write your own content here.]

4.6.3 Verification and human accountability

[GUIDE NOTE — delete before submission] One paragraph. Every factual claim verified against original sources, every cited paper read, all assisted code run and inspected, evidence retained.

[Write your own content here.]

4.6.4 Privacy, confidentiality and intellectual property

[GUIDE NOTE — delete before submission] One paragraph. What must never be entered into a tool, and how minimal synthetic examples are used instead.

[Write your own content here.]

4.6.5 Transparency, disclosure and AI-use register

[GUIDE NOTE — delete before submission] Summarise here; keep the full register in Appendix L with tool, date, purpose, input category, verification, and whether the output was retained, rejected or rewritten.

[Write your own content here.]

4.6.6 Generative AI declaration

[GUIDE NOTE — delete before submission] The declaration statement itself, consistent with the front-matter disclosure.

[Write your own content here.]

4.6.7 Project-specific Generative AI risk assessment

[GUIDE NOTE — delete before submission] Table: risk | possible consequence | control and evidence.

[Write your own content here.]

4.7 Chapter summary

[GUIDE NOTE — delete before submission] One paragraph. Do not claim all risks have been eliminated.

[Write your own content here.]

REFERENCES

[GUIDE NOTE — delete before submission] Harvard format, consistently applied. Only sources cited in the text. Verify every author, year, title, venue and DOI against the original publication. Not counted in the 35-page limit.

[Write your own content here.]

APPENDICES

[GUIDE NOTE — delete before submission] Every appendix needs a title and must be referred to from the main text. Essential arguments, the gap justification, the core architecture and the evaluation design must remain in the main body, not here. Delete any appendix that does not apply to your project.

Appendix A: Detailed search strategy and screening log

[Write your own content here.]

Appendix B: Concept map

[Write your own content here.]

Appendix C: Literature survey matrix

[Write your own content here.]

Appendix D: Research gap analysis matrix

[Write your own content here.]

Appendix E: Dataset comparison matrix and data dictionary

[Write your own content here.]

Appendix F: Stakeholder onion, elicitation instruments and evidence

[Write your own content here.]

Appendix G: Full functional and non-functional requirements

[Write your own content here.]

Appendix H: Use cases, use-case descriptions and activity/sequence diagrams

[Write your own content here.]

Appendix I: Detailed architecture and data-flow diagrams

[Write your own content here.]

Appendix J: Detailed Gantt chart, backlog and milestone plan

[Write your own content here.]

Appendix K: Full risk register and issue log

[Write your own content here.]

Appendix L: Generative AI use declaration and AI-use register

[Write your own content here.]

Appendix M: Preliminary proof-of-concept screenshots, test logs and sample outputs

[Write your own content here.]

Appendix N: Ethics clearance material and data-management approvals

[Write your own content here.]